

CS305: Computer Networks

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Koteswararao Kondepu

k.kondepu@iitdh.ac.in

OSPF routing - Recap

Which OSPF area is mandatory?

- A. Area 1
- B. Area 0
- C. Area 10
- D. Backbone not required

Which router is elected in a multi-access network?

- A. Only DR
- B. Only BDR
- C. DR and BDR
- D. None

What is the default Hello interval (broadcast network)?

- A. 5 seconds
- B. 10 seconds
- C. 30 seconds
- D. 40 second

Two OSPF routers are not forming adjacency. What is MOST likely cause?

- A. Same area
- B. Matching Hello timers
- C. Different subnet
- D. Same authentication

What is the default Dead interval?

- A. 5 seconds
- B. 10 seconds
- C. 30 seconds
- D. 40 second

OSPF metric is based on:

- A. Delay
- B. Bandwidth
- C. Hop count
- D. Load

OSPF routing - Recap

Which router is elected in a multi-access network?

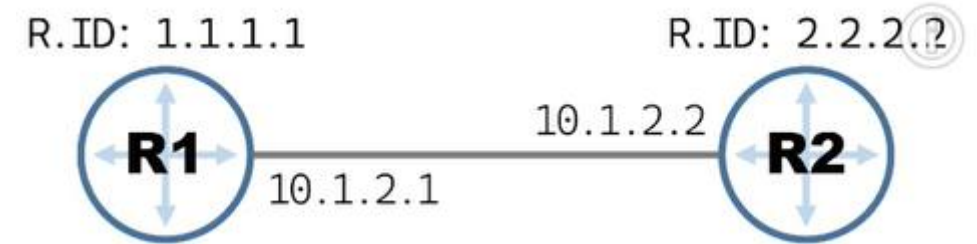
- A. Only DR
- B. Only BDR
- C. DR and BDR
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Two OSPF routers are not forming adjacency. What is MOST likely cause?

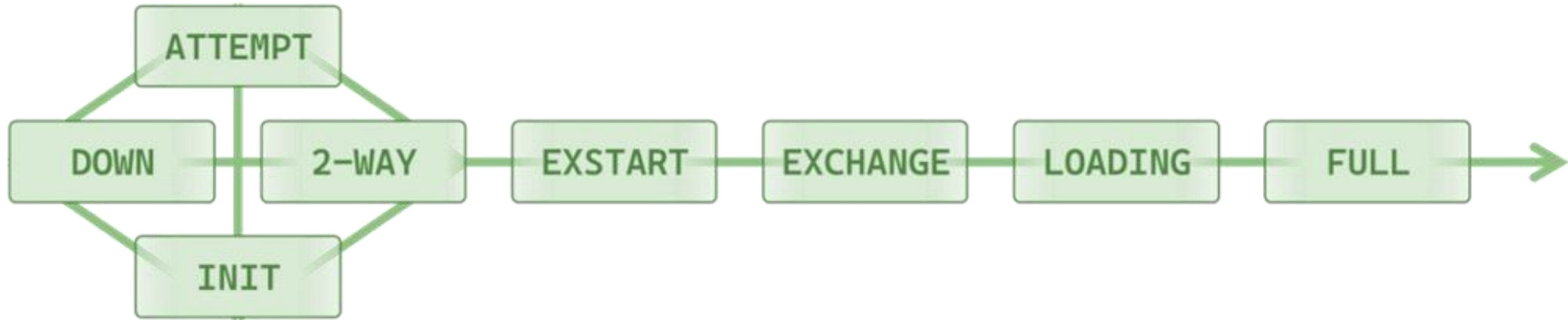
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OSPF metric is based on:

- A. Delay
- B. Bandwidth
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- D. Load



OSPF Adjacency (1/5)



State: DOWN

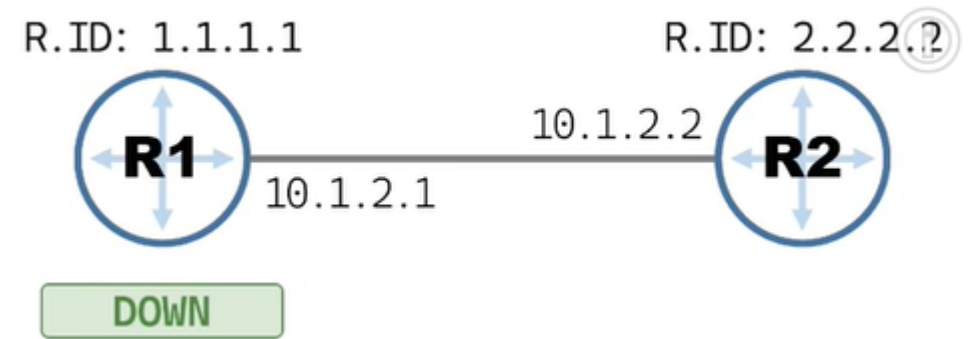
- Initial state when OSPF first configure
- Sending periodic hello to 224.0.0.5

State: ATTEMPT

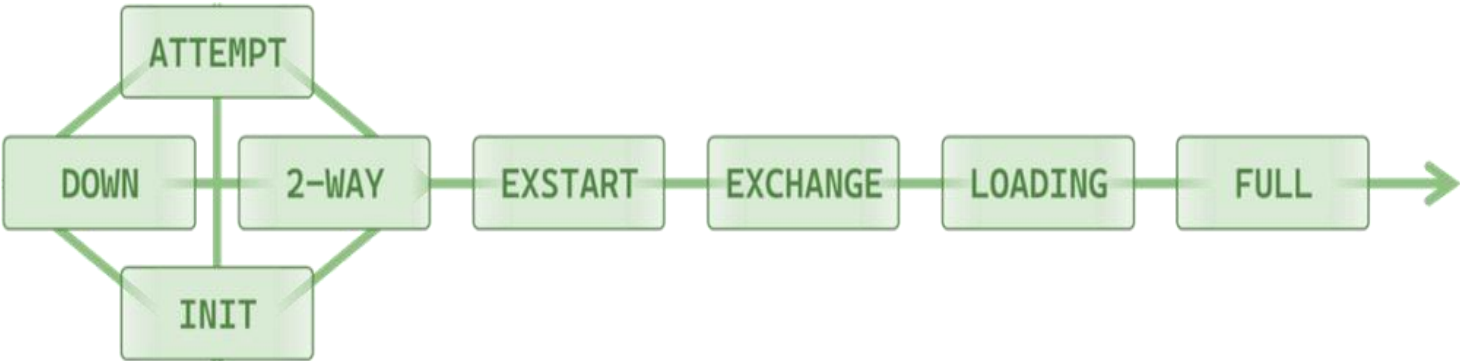
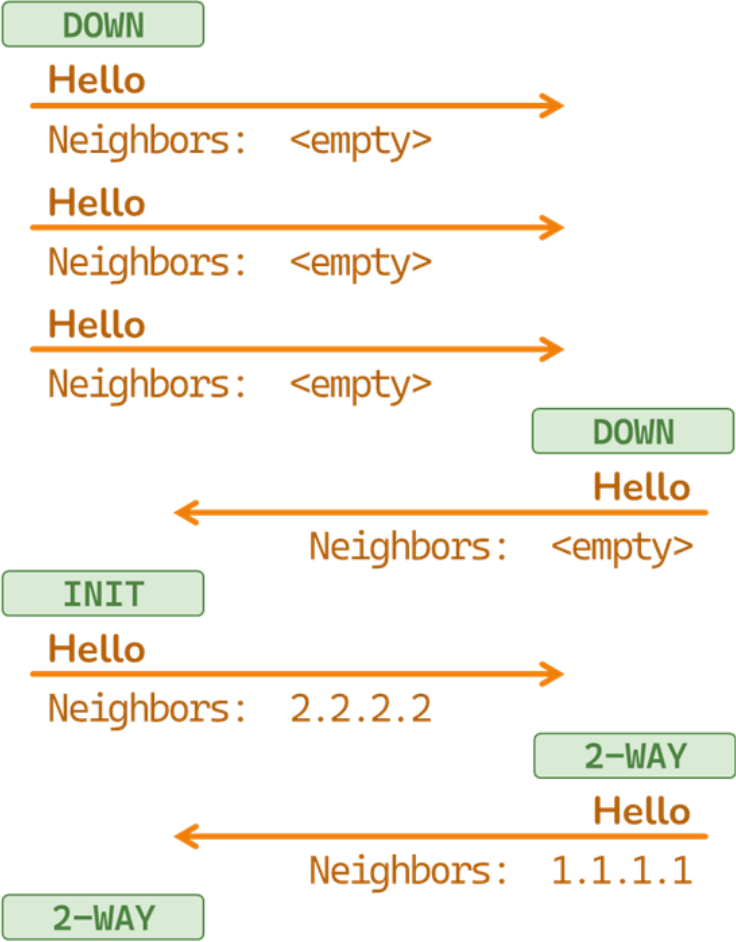
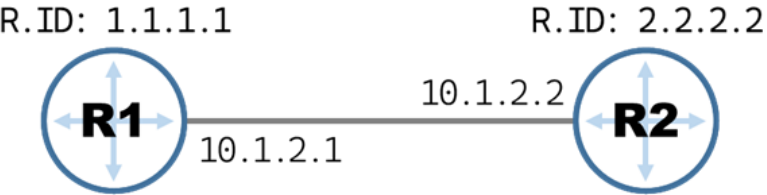
- Manually configuration / sending unicast hellos

State: INIT

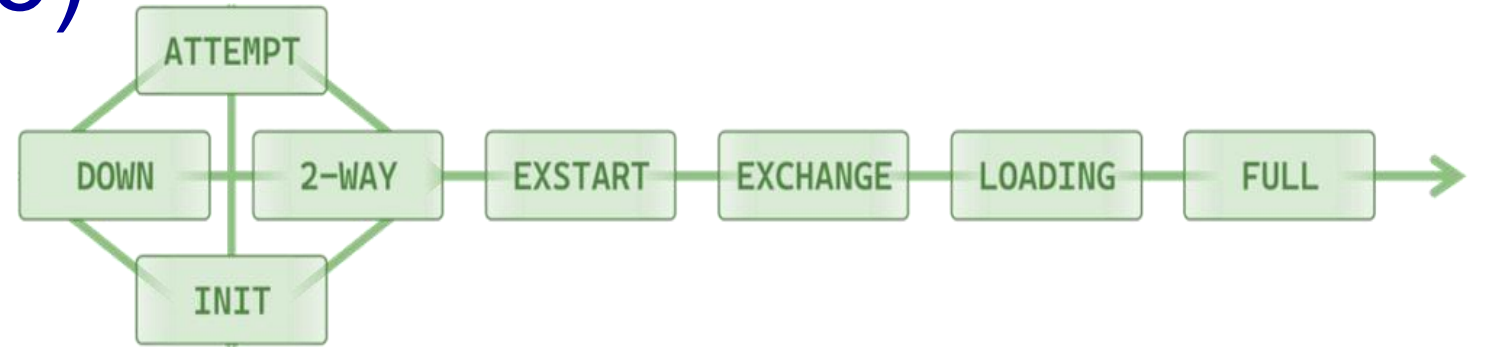
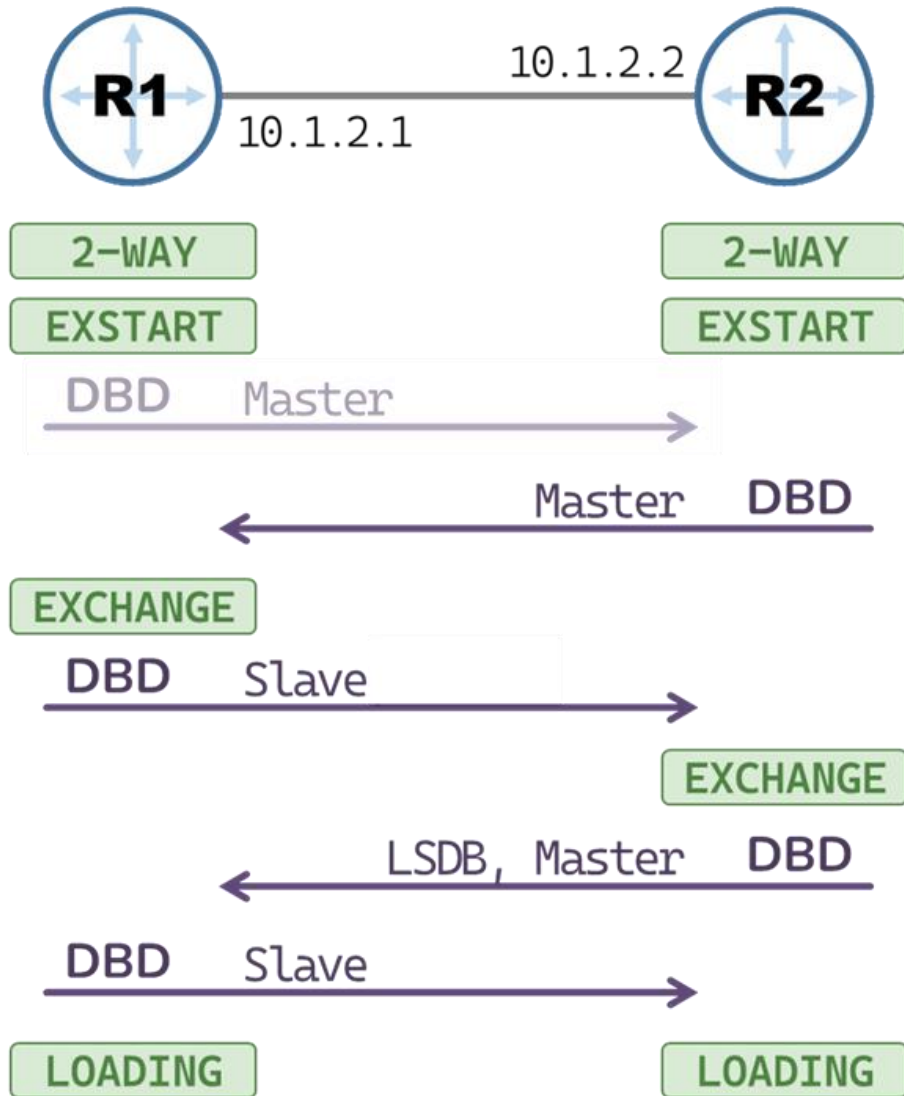
- Received a hello packet



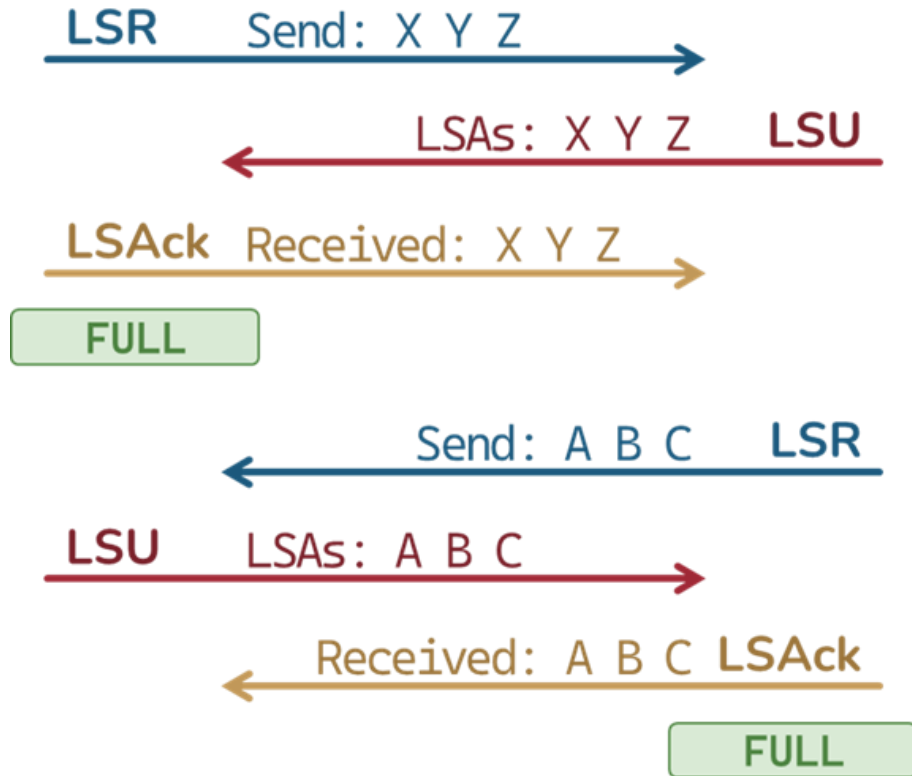
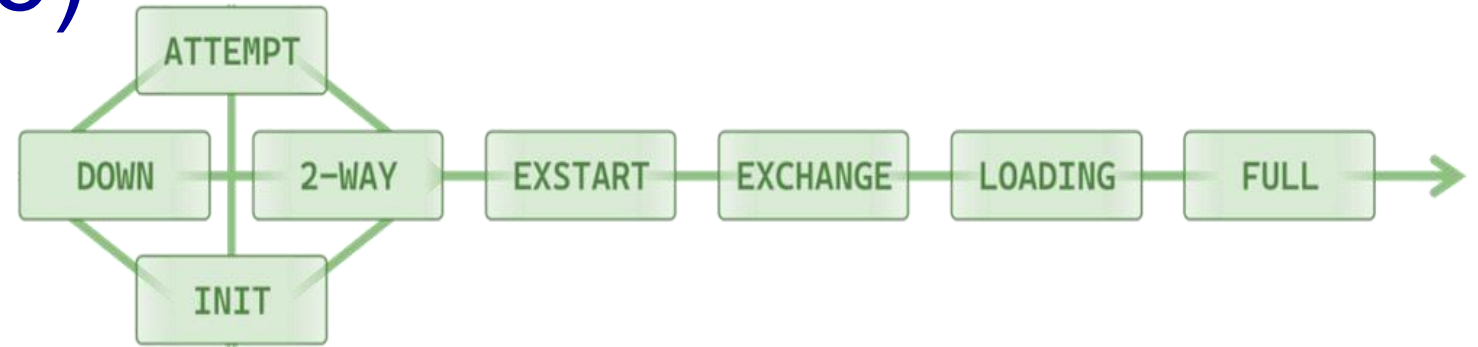
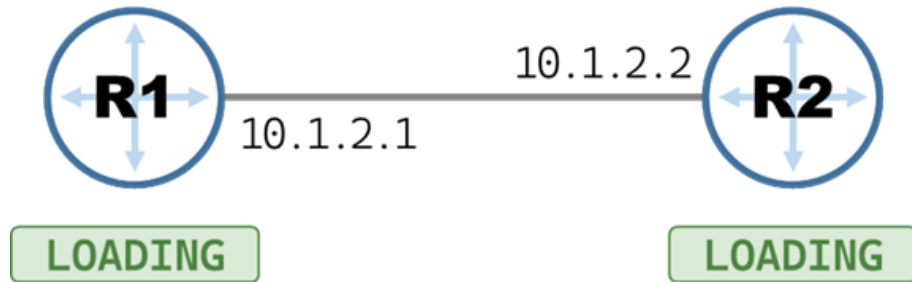
OSPF Adjacency (2/5)



OSPF Adjacency (3/5)



OSPF Adjacency (4/5)



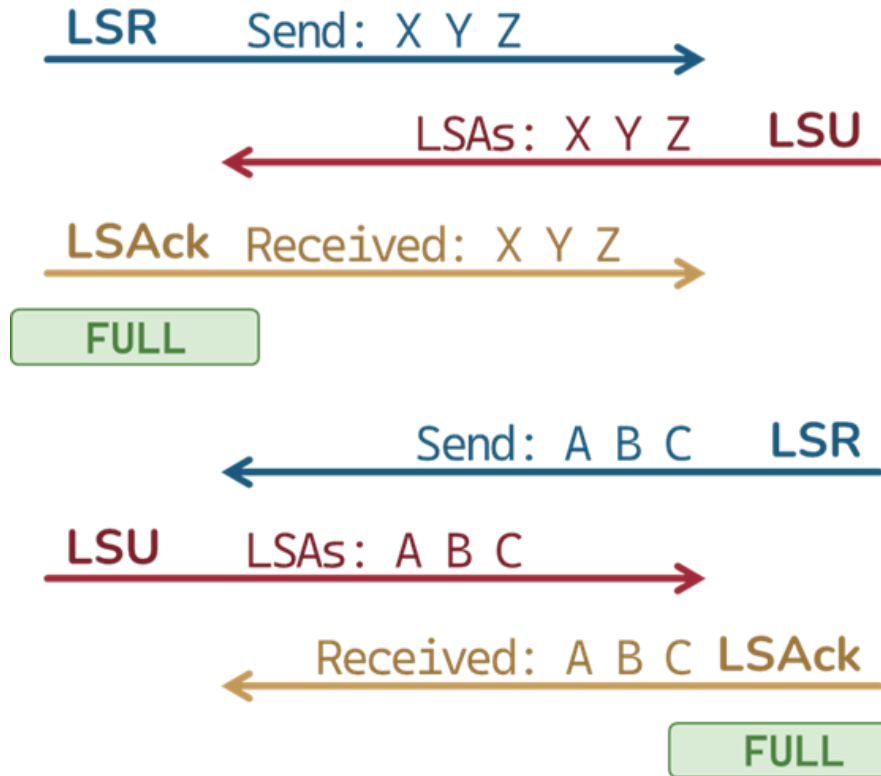
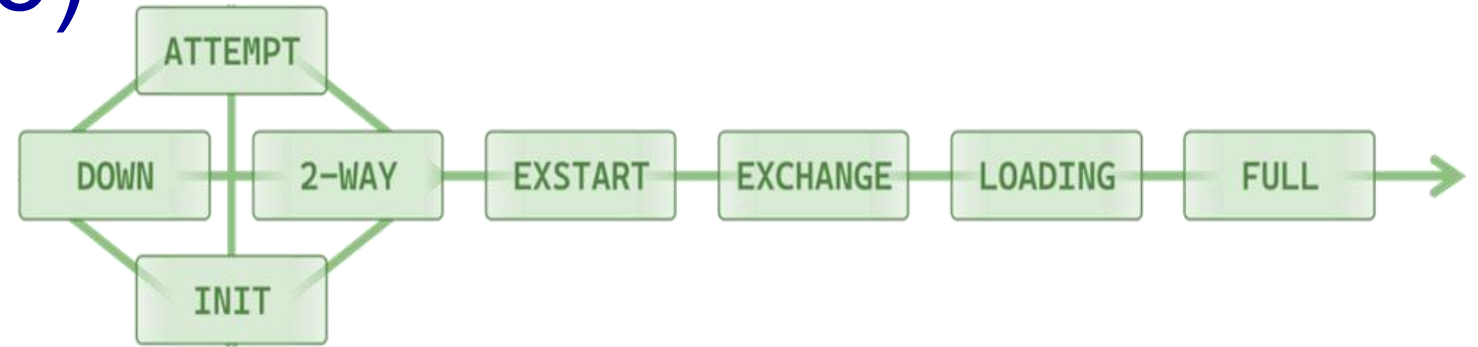
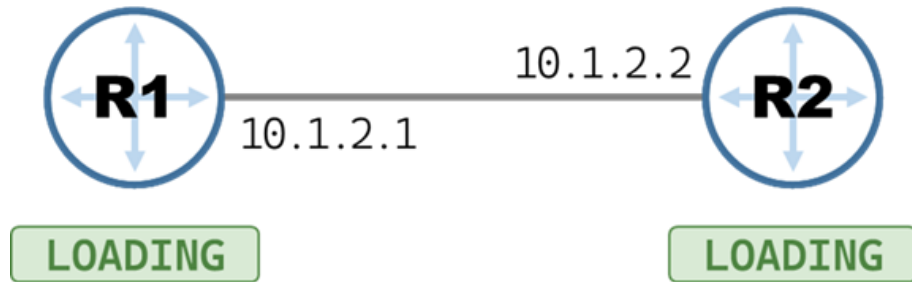
What are OSPF neighbor states?

Down → Init → 2-Way → ExStart → Exchange → Loading → Full

What is required for OSPF routers to form adjacency?

- A. Same IP address
- B. Same area ID
- C. Same hostname
- D. Same MAC address

OSPF Adjacency (5/5)



What are OSPF neighbor states?

Down → Init → 2-Way → ExStart → Exchange → Loading → Full

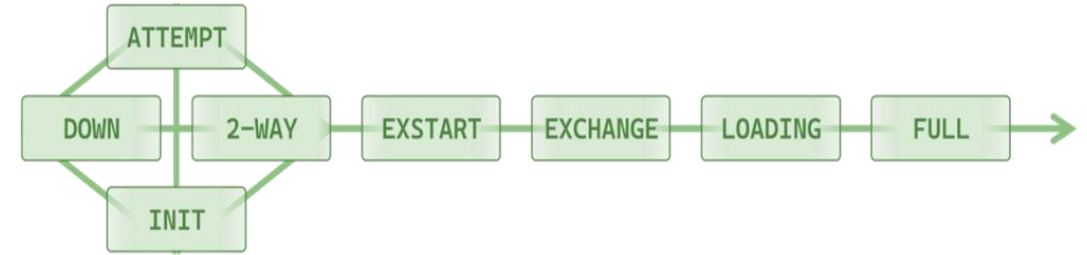
What is required for OSPF routers to form adjacency?

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- B. Same area ID
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- D. Same MAC address

OSPF Adjacency – Questions?

What is required for OSPF routers to form adjacency?

- A. Same IP address
- B. Same area ID
- C. Same hostname
- D. Same MAC address



Which state indicates that routers have become neighbors but not fully adjacent?

- A. Full
- B. Exchange
- C. 2-Way
- D. Loading

In which state do routers exchange database summaries?

- A. Init
- B. ExStart
- C. Exchange
- D. Loading

What is the first state in OSPF neighbor formation?

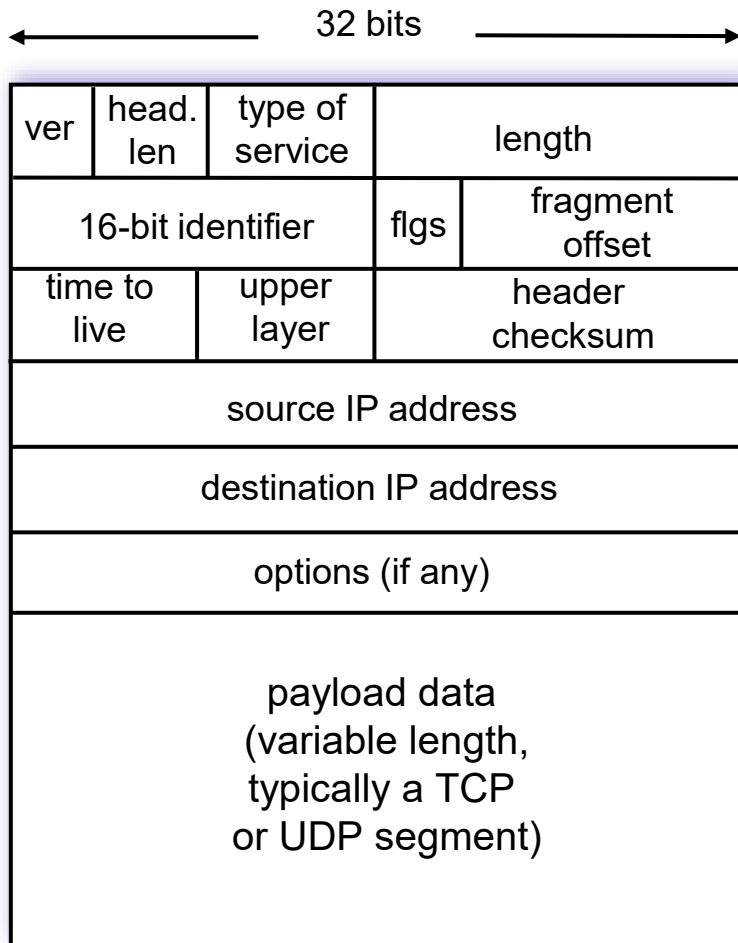
- A. Init
- B. Down
- C. 2-Way
- D. Exchange

What are OSPF neighbor states?

Down → Init → 2-Way → ExStart → Exchange → Loading → Full

Network layer: “control plane” roadmap

■ Internet Control Message Protocol



1. Which of the following is NOT provided by IP?

- A. Logical addressing
- B. Packet forwarding
- C. Guaranteed delivery
- D. Routing

IP is considered a:

- A. Connection-oriented protocol
- B. Reliable protocol
- C. Connectionless and no reliable protocol
- D. Secure protocol

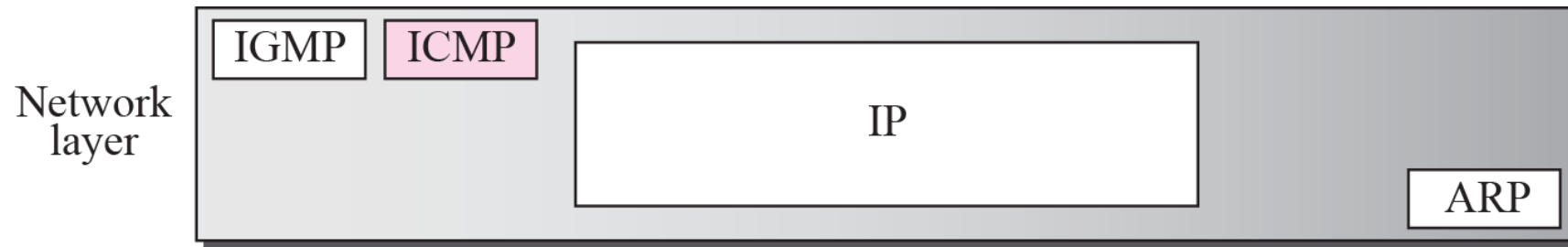
3. Which feature is missing in IP that is provided by TCP?

- A. Addressing
- B. Routing
- C. Error recovery
- D. Packet forwarding



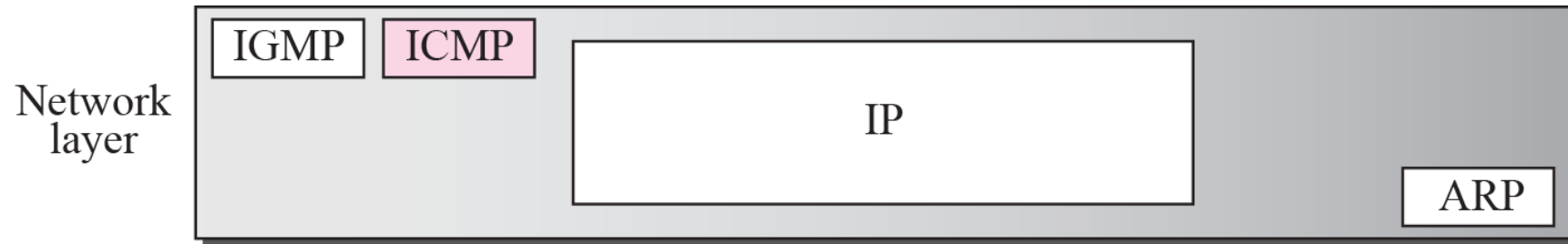
ICMP: Introduction

- The IP protocol has no error-reporting or error correcting mechanism.
- What happens if something goes wrong? What happens if a router must discard a datagram because it cannot find a router to the final destination, or because the time-to-live field has a zero value?
- These are examples of situations where an error has occurred and the IP protocol has no built-in mechanism to notify the original host.

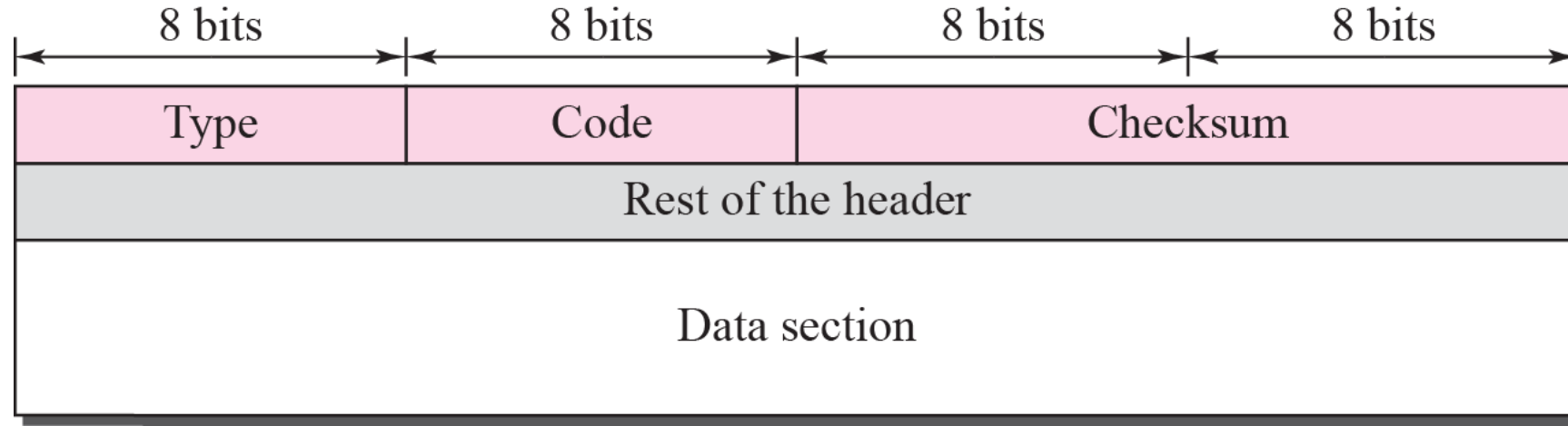


ICMP: Messages

- ICMP messages are divided into two broad categories:
 - error-reporting messages
 - query messages.
- The error-reporting messages report problems that a router or a host (destination) may encounter when it processes an IP packet.
- The query messages, which occur in pairs, help a host or a network manager get specific information from a router or another host.



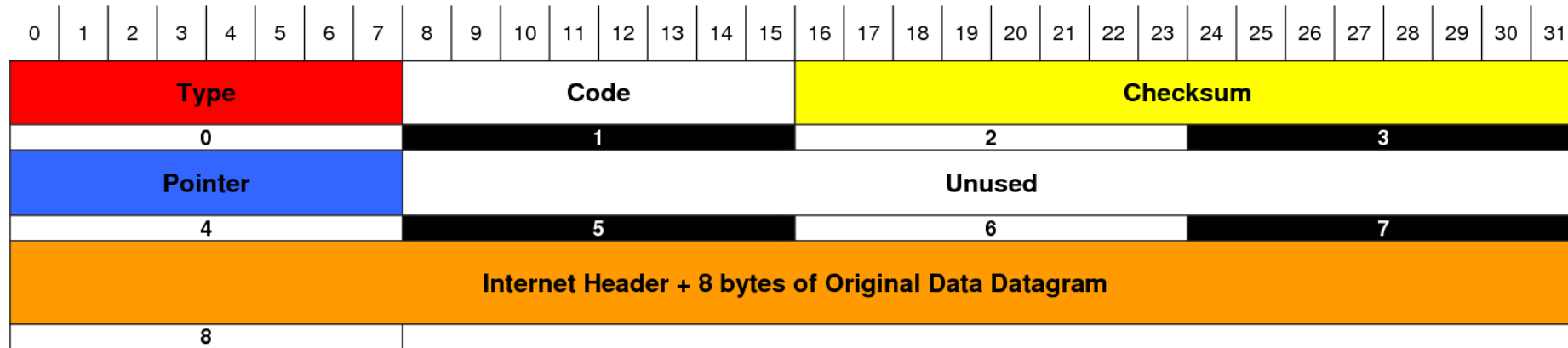
ICMP: Packet Format, Messages



Category	Type	Message
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply

ICMP: Packet Format, Messages

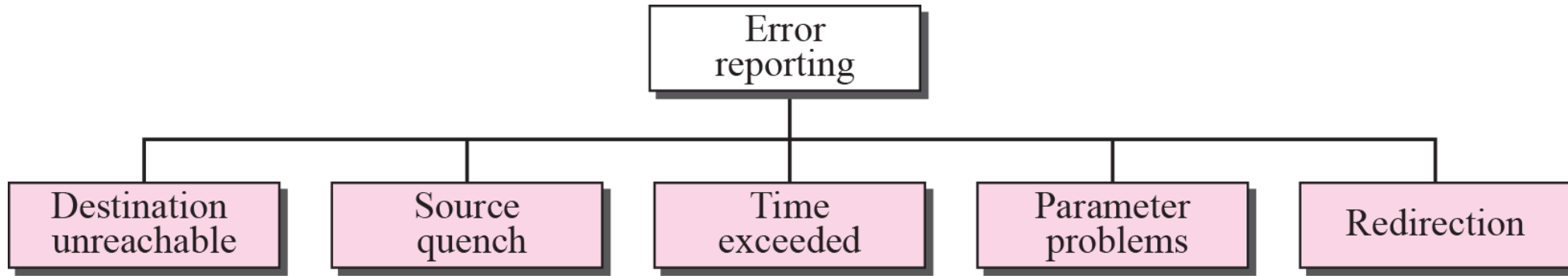
ICMP Parameter Message Format



Type	Code	Meaning
0	0	Echo Reply
3	0	Net Unreachable
	1	Host Unreachable
	2	Protocol Unreachable
	3	Port Unreachable
	4	Frag needed and DF set
	5	Source route failed
	6	Dest network unknown
	7	Dest host unknown
	8	Source host isolated
	9	Network admin prohibited
	10	Host admin prohibited
	11	Network unreachable for TOS
	12	Host unreachable for TOS
	13	Communication admin prohibited
4	0	Source Quench (Slow down/Shut up)

Type	Code	Meaning
5	0	Redirect datagram for the network
	1	Redirect datagram for the host
	2	Redirect datagram for the TOS & Network
	3	Redirect datagram for the TOS & Host
8	0	Echo
9	0	Router advertisement
10	0	Router selection
11	0	Time To Live exceeded in transit
	1	Fragment reassemble time exceeded
12	0	Pointer indicates the error (Parameter Problem)
	1	Missing a required option (Parameter Problem)
	2	Bad length (Parameter Problem)
13	0	Time Stamp
14	0	Time Stamp Reply
15	0	Information Request
16	0	Informaiton Reply
17	0	Address Mask Request
18	0	Address Mask Reply
30	0	Traceroute (Tracert)

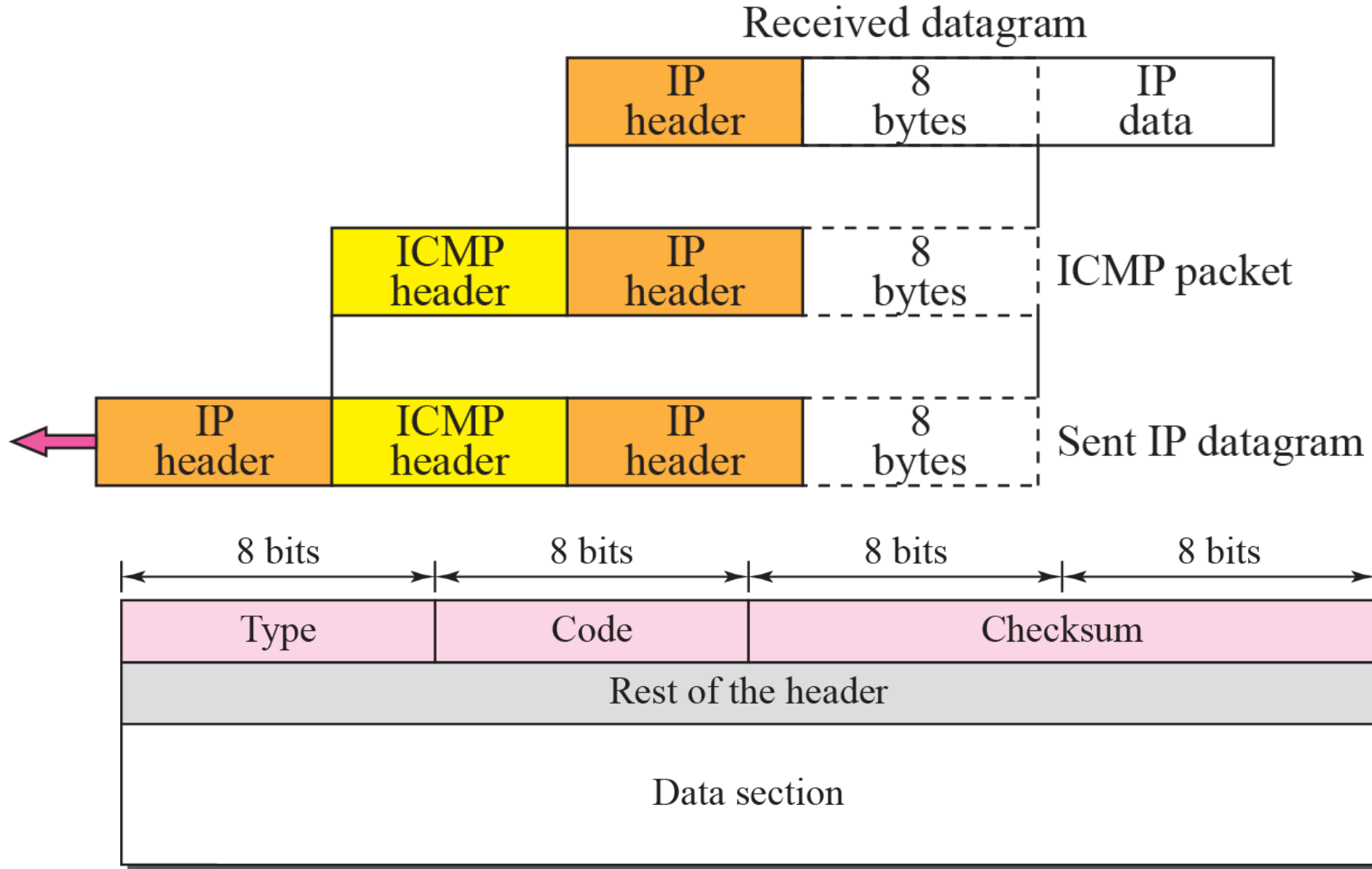
ICMP: Error Report Messages



Type	Code	Meaning
0	0	Echo Reply
3	0	Net Unreachable
	1	Host Unreachable
	2	Protocol Unreachable
	3	Port Unreachable
	4	Frag needed and DF set
	5	Source route failed
	6	Dest network unknown
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ICMP: Content of data field for the error message



ICMP: Destination Unreachable

Type: 3	Code: 0 to 15	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

Type	Code	Meaning
0	0	Echo Reply
3	0	Net Unreachable
	1	Host Unreachable
	2	Protocol Unreachable
	3	Port Unreachable
	4	Frag needed and DF set
	5	Source route failed
	6	Dest network unknown
	7	Dest host unknown
	8	Source host isolated
	9	Network admin prohibited
	10	Host admin prohibited
	11	Network unreachable for TOS
	12	Host unreachable for TOS
	13	Communication admin prohibited
4	0	Source Quench (Slow down/Shut up)

- Destination-unreachable messages with codes 2 or 3 can be created only by the destination host.*
- Other destination-unreachable messages can be created only by routers.*

ICMP: Source Quench

Type: 4	Code: 0	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

Type	Code	Meaning
0	0	Echo Reply
3	0	Net Unreachable
	1	Host Unreachable
	2	Protocol Unreachable
	3	Port Unreachable
	4	Frag needed and DF set
	5	Source route failed
	6	Dest network unknown
	7	Dest host unknown
	8	Source host isolated
	9	Network admin prohibited
	10	Host admin prohibited
	11	Network unreachable for TOS
	12	Host unreachable for TOS
	13	Communication admin prohibited
4	0	Source Quench (Slow down/Shut up)

- A source-quench message informs the source that a datagram has been discarded due to congestion in a router or the destination host.*
- The source must slow down the sending of datagrams until the congestion is relieved.*
- One source-quench message is sent for each datagram that is discarded due to congestion.*

thank you!

email:

k.kondepu@iitdh.ac.in